

Anshika Shukla

anshikashukla0322@gmail.com | +91-8957631665 | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

ECE undergraduate with experience in software development, AI/ML, and data analysis. Proficient in Python, React, FastAPI, and SQL, with hands-on experience building data pipelines, POC prototypes, and cloud-backed dashboards. Adept at translating analytical insights into digital solutions across cross-functional teams, with strong documentation, prototyping, and collaborative delivery skills.

EDUCATION

Banasthali Vidyapith — B.Tech, Electronics & Communication Engineering CGPA: 7.25/10 July 2023 – May 2027

- Preprint Research ([SSRN ID: 6763558](#)): "Advancements in Brain-Computer Interfaces for Communication Disorders and Assistive Technologies"
- Poster Presentation — WACIIS 2025 International Conference

KEY ACHIEVEMENTS

- 2nd Runner-Up — HACKVERSE (National, 30-hr, 100+ teams): Built SafeGaze achieving 95% detection accuracy under live competition conditions.
- Top 10 — HACKCELESTIA (National): Architected SatLink edge-AI platform with satellite relay delivering alerts in <10s in zero-connectivity zones.
- Selected — Sanchar Saathi Initiative, Govt. of India (DoT): Recognized nationally for digital safety and telecommunications innovation.
- GSSoC 2026 Contributor — Girl Script Summer of Code: Selected as open-source contributor for the 2026 cohort.

EXPERIENCE

Technical Workshop Lead / Core Member — SAMARTHYA Electronics Club Sept 2024 – Present

Banasthali Vidyapith, Jaipur, Rajasthan

- Boosted project completion by 40% and lab participation by 35% by designing 5+ IoT/embedded workshops for 200+ students.
- Accelerated 20+ junior students' competition readiness mentoring circuit design and sensor integration at national hackathons.

PROJECTS

Flight-Sense-AI — Real-Time Anomaly Detection | Python · Isolation Forest · OpenSky ADS-B · React · Supabase

- Engineered ML anomaly detection pipeline (Isolation Forest) on live ADS-B RF telemetry streams over India; flagged irregular flight patterns with <200ms latency.
- Built React dashboard with real-time signal ingestion via WebSockets and Supabase; processed 1,000+ concurrent aircraft data points.

May 2026

SMART-GUARD — IoT Smart Helmet | Arduino Uno · C++ · MPU6050 · GSM · Twilio API

- Reduced emergency response time to <2 sec by engineering 4-sensor fusion (MPU6050, IR, MQ3, UV) for accident detection with GSM/Twilio SMS+GPS alerts.
- Developed alert logic and Twilio API notification pipeline; documented sensor thresholds and system behavior for cross-team validation.

Apr 2026

SatLink — Edge-AI Emergency Response Platform | Python · TensorFlow Lite · YOLO · OpenCV · Fourier Analysis · Twilio API · Flask

- Achieved 90%+ fire/flood classification accuracy via Fourier and Wavelet feature extraction on real-time camera feeds.
- Built SOS pipeline integrating GPS tracking, satellite relay, and Twilio SMS/Call alerts; sub-10s notification in low-connectivity zones.

Jan 2026

Real-Time Satellite Safety & Health Dashboard | YOLO-Tiny · Python · MQTT · WebSockets · Edge AI

- Automated anomaly detection using YOLO-Tiny achieving 95%+ accuracy; reduced manual monitoring effort by 60% across live visual data feeds.
- Engineered MQTT/WebSocket telemetry stack processing 1,000+ sensor updates/sec at 180ms latency for continuous health monitoring.

Oct 2025

SignEase — Real-Time ASL Recognition | Python · TensorFlow Lite · OpenCV · Raspberry Pi · CNN

- Compressed CNN 81% (45 MB to 8.5 MB) with 94% accuracy at <100ms on Raspberry Pi; applied quantization-aware training for embedded inference.
- Trained on 80K+ images with 18% robustness gain via augmentation; implemented real-time signal preprocessing pipeline with OpenCV.

Sept 2025

TECHNICAL SKILLS

Signal Processing: Digital Filtering · Fourier/Wavelet Analysis · Sensor Fusion · Feature Extraction · Anomaly Detection · Time-Series Analysis

AI/ML & Deep Learning: TensorFlow · TensorFlow Lite · Keras · PyTorch (fundamentals) · Scikit-learn · CNNs · RNNs · Model Quantization · Inference Optimization

Programming: Python (NumPy, SciPy, OpenCV) · C/C++ · MATLAB · JavaScript · SQL

Hardware & Embedded: Raspberry Pi · Arduino Uno · MPU6050 · GSM/GPS Modules · Embedded Sensors · MQTT · WebSockets

Tools & Platforms: Git · GitHub · VS Code · Jupyter · Streamlit · FastAPI · Google Cloud APIs

CERTIFICATIONS

- Building Data & AI Solutions on Google Cloud — 365 Data Science (100%, March 2026)
- NASA ARSET | MATLAB Onramp — MathWorks | Career Essentials in Generative AI — Microsoft & LinkedIn
- Drone Certification — Dept. of AI, Banasthali Vidyapith

SPOKEN LANGUAGES

English (Proficient) · Hindi (Native) · French (B1 Diploma)